

When Predictive Surrogates Fail as RL Environments: A Calibrated Physical Twin as Soft Regularizer for HVAC Control

May 28, 2026

This document contains the eleven supplementary tables (S1–S11) referenced in the main manuscript. They cover the Hou-and-Evins *Reporting Level 3* requirements for surrogate-development disclosure: sample generation and sizing, preprocessing, feature significance, input independence, split representativeness, scaling, training hyperparameters, architecture justification, targeted sensitivity, and replicative/predictive validity.

Contents

Origin, parameter ranges, and physical justification of each transition corpus.

dataset	source_files	episodes	step_group	controllers	season	policy	combination	nonex	maxmb	minmb	powerc	npio
v3_hourlydata	dir\{tsingto	v26\}	boptes36002	tsmpolyreat,autochras,bing	580	summer	38.828	inter-24.4	35.0	0.0	155	

[illegible]

```
v35_collect_data/Block18342/lorazepam_2000/boptestoll_hack_14int15min1string_141815578862669312858312434.399999999999999981721
24.400000000000000006
```

Numerical evidence for the chosen corpus sizes including learning-curve checks.

Table 2: Table S2. Sample-size justification.

dataset_id	rows	episodes	steps	new_bots	primary_collection	in_training	in_validation	in_test	secondary	used	secondary	in_validation	in_test
v3_hourly	51200	tsup	16	3600		thermostatic	0.083596	0.368982	0.25084	static	345	23323	11136738

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dataset_id	rows	episodes	steps	groups	new_bootstrap	optimal	collective	performance	main	secondary	value	secondary	prediction
v35_prepared	10745	min_bootstrap	900		0.0	calibrated_rollout	0.64407234617085	calibrated_rollout	0.2818961573295974				
collected_154884	exploration	900			17.181544331709542	use_c	1.1145992687408475	safe	3.8445638693675073	as			

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dataset_id	rows	episodes	steps	new_bots	test_episodes	test_steps	test_rmse_c	test_rmse_s	test_rmse_a	test_rmse_d	test_rmse_e	test_rmse_f	test_rmse_g	test_rmse_h	test_rmse_i	test_rmse_j	test_rmse_k	test_rmse_l	test_rmse_m	test_rmse_n	test_rmse_o	test_rmse_p	test_rmse_q	test_rmse_r	test_rmse_s	test_rmse_t	test_rmse_u	test_rmse_v	test_rmse_w	test_rmse_x	test_rmse_y	test_rmse_z
collected_15	21192	12	900	9.483321352799738	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765	0.9210515043790765		

Table S3: Stage A telemetry preprocessing.

Latency compensation, bias removal, normalisation, denoise, and causal delta.

Table 3: Table S3. Stage A telemetry preprocessing.

stage_a_operation	implementation	parameter_or_rule	selection_criterion	purpose
latency compensation search	<code>_undo_delay_array</code> over candidate lags	lag in $[0, \text{max_latency_search}]$	choose lag minimizing surrogate one-step RMSE against observed next-step temperature	undo thermostat/telemetry delay before calibration
temperature bias removal	median residual off-set subtraction	$\text{bias_est} = \text{median}(t_{\text{obs_next}} - t_{\text{pred}})$	center residual distribution before Stage B/C calibration	avoid fitting static sensor bias as building physics
power affine normalization	least-squares fit $p_{\text{obs}} \setminus \text{approx scale} * p_{\text{pred}} + \text{bias}$	$\text{scale_est}, \text{bias_p_est}$ from linear solve	reduce power-channel mismatch before nonlinear head refinement	separate systematic meter scaling from dynamic error
rolling denoise	rolling median + exponential moving average	$\text{smooth_window} = 5$	suppress high-frequency telemetry noise without changing regime identity	stabilize downstream inverse calibration
causal delta recomputation	$\text{delta_t} = t_{\text{zone_next}} - t_{\text{zone}}$ after preprocessing	derived from corrected current and next temperature	preserve state-transition consistency after artifact correction	avoid propagating artifact-driven temperature-history features

Table S4: Feature significance.

Per-feature retention rationale with quantitative selection criteria.

Table 4: Table S4. Feature significance.

variant	obs_able	alpha	feature	power	history	feature	importance	peak	violation	peak	importance	peak	importance	peak	importance
with_power	raw	raw	raw	raw	raw	1.269802618399051108398814128586285238095231	2.0	2.0							
no_powemo	power	raw	raw	raw	raw	1.2814149612895908399705857342857095238095231	2.0	2.0							
no_delta	no_delta	raw	raw	raw	raw	1.0360754011876097400725376839587085714285711	2.001232623987481								
no_powemo	no_delta	raw	raw	raw	raw	1.0622488272307337768309384764023809523811	2.0	2.0							
no_prev	no_prev	raw	raw	raw	raw	1.2587796902705528874854566366755952380952381	2.0	2.0							
no_temp	no_history	raw	raw	raw	raw	1.084782460928592820439971483856091761904771	2.002767727007382								
causal	smooth	causal	smooth	raw	raw	1.12878876978605781042083533337341	1	2.0	2.0						
no_delta	no_delta	clipped_log	clipped_log	raw	raw	1.089985241123958728732755952380952381	2.0	2.0000053							
no_delta	no_delta	clipped_log	clipped_log	raw	raw	1.046450511601282977608355732336671128571431	2.0	2.0141638							

Table S5: Input independence.

Pearson and mutual-information independence checks across input channels.

Table 5: Table S5. Input independence.

feature_i	feature_j	pearson_r	abs_pearson	normalized_mutual	source_file	interpretation
t_zone	t_amb	-0.14513070212380655	0.14513070212380655	0.005235472589968666	data/block_1_2lowurto	to
t_zone	hour_sin	-0.3633857830047388	0.3633857830047388	0.005804753664319774	data/block_1_2lowurto	to
t_zone	hour_cos	-0.0029075796864029397	0.0029075796864029397	0.0059213489918670795	data/block_1_2lowurto	to
t_zone	day_sin	-0.14385882436029465	0.14385882436029465	0.0050135677674280744	data/block_1_2lowurto	to
t_zone	day_cos	0.11199728742103762	0.11199728742103762	0.0050914299627210744	data/block_1_2lowurto	to
t_zone	prev_action_a0	0.130331087242927136	0.130331087242927136	0.006303792243196744	data/block_1_2lowurto	to
t_zone	prev_action_a1	0.148177994054100518	0.148177994054100518	0.0057394987142150346	data/block_1_2lowurto	to
t_zone	power_history_proxy	0.244638749845961251	0.244638749845961251	0.005005953794486244	data/block_1_2lowurto	to
t_zone	delta_t_history	-0.17908344379114244	0.17908344379114244	0.0042423307333590736	data/block_1_2lowurto	to
t_amb	hour_sin	-0.3075536006017538	0.3075536006017538	0.00383963151828880744	data/block_1_2lowurto	to
t_amb	hour_cos	-0.3656114451793563	0.3656114451793563	0.004318018105920084	data/block_1_2lowurto	to
t_amb	day_sin	0.024585699176605825	0.024585699176605825	0.0032213442130890724	data/block_1_2lowurto	to
t_amb	day_cos	0.013184426544837684	0.013184426544837684	0.003765185451534816924	data/block_1_2lowurto	to
t_amb	prev_action_a0	-0.8005295659104322	0.8005295659104322	0.00227795942535390884	data/block_1_2msurto	to
t_amb	prev_action_a1	-0.057710407906264465	0.057710407906264465	0.0042544721945466744	data/block_1_2lowurto	to
t_amb	power_history_proxy	0.638125673837222	0.638125673837222	0.00226740679512406734	data/block_1_2msurto	to
t_amb	delta_t_history	0.06085142314870052	0.06085142314870052	0.002020249825670774	data/block_1_2lowurto	to

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feature_i	feature_j	pearson_r	abs_pearson_normalized_mutual_info	source_file	interpretation
hour_sin	hour_cos	-	9.7262464858750073e-05	data/block_1_2msdcoraga	data/block_1_2msdcoraga
hour_sin	day_sin	-	0.012496678250250352	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_sin	day_cos	0.012387244067789052	0.012387244067789052	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_sin	prev_action_a0	0.4759432765580074	0.4759432765580074	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_sin	prev_action_a1	-	0.04595594325102461	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_sin	power_history_prev	0.4276574605727682	0.4276574605727682	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_sin	delta_t_history	0.0024932398394025	0.0024932398394025	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_cos	day_sin	0.000810706136756058	0.000810706136756058	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_cos	day_cos	-	0.0008478628403947587	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_cos	prev_action_a0	0.548550822692035	0.548550822692035	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_cos	prev_action_a1	-	0.22187703800857933	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_cos	power_history_prev	0.2514582620463795	0.2514582620463795	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
hour_cos	delta_t_history	-	0.0748145754637328	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_sin	day_cos	-	0.9857151177566247	data/block_1_2highroaga	data/block_1_2highroaga
day_sin	prev_action_a0	0.0614371364175635	0.0614371364175635	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_sin	prev_action_a1	0.0087171699338308	0.0087171699338308	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_sin	power_history_prev	0.0521809527737795	0.0521809527737795	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_sin	delta_t_history	0.0026841852551630	0.0026841852551630	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_cos	prev_action_a0	-	0.07642267373096458	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_cos	prev_action_a1	-	0.02245908931409531	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_cos	power_history_proxy	0.06929017194900049	0.06929017194900049	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
day_cos	delta_t_history	-	0.002730305666116105	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
prev_action_a0	prev_action_a1	-	0.11254131048809551	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
prev_action_a0	power_history_prev	0.7437088684650934	0.7437088684650934	data/block_1_2msdcoraga	data/block_1_2msdcoraga
prev_action_a0	delta_t_history	-	0.1383599349888946	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
prev_action_a1	power_history_prev	0.3881301312121038	0.3881301312121038	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
prev_action_a1	delta_t_history	0.0265869600260782	0.0265869600260782	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga
power_history_prev	delta_t_history	-	0.09333818820645873	data/block_1_2lowurtoaga	data/block_1_2lowurtoaga

Table S6: Train/validation/test split representativeness.

Distributional checks between corpus partitions.

Table 6: Table S6. Train/validation/test split representativeness.

pipeline	split_mode	train_rows	val_rows	test_definition	train_group	validation_group	coverage	seasons	weather	policy
v3_hourly	random	40666	10840	external 15-minute BOPTEST benchmark scenarios	13.0	4.0	autumn, spring, summer, winter	cool, heat, mixed, heat, mild		
v35_inverse	episodic	185800	12686	only preparation rollout validation plus live thermostat BOPTEST transfer benchmark	6.0	2.0	peak_heat, peak_cool, peak_heat_cool, peak_cool_cool, peak_cool_heat, peak_heat_heat			

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pipeline	split_mode	train_rows	val_rows	test_definition	train_group	val_group	aggr	over	sagittal_sensor	train_policy	val_policy
hybrid_thermo_mechanical	dataset	split		two unseen live BOPTTEST heating sce- narios: peak_heat_window and typi- cal_heat_window							e

Standardisation parameters reused at inference time.

variable	context	scaling_method	parameters	source_file	justification
surrogate_t_zone	v3/v3.5 surrogate input	affine min-max to [-1, 1]	min=15 C; max=35 C	surrogate/rc_node_v35.py	Represents the learned thermal state within the training support used by the direct-TSup surrogate.
surrogate_t_amb	v3/v3.5 surrogate input	affine min-max to [-1, 1]	min=-10 C; max=40 C	surrogate/rc_node_v35.py	Normalizes outdoor-air temperature for the heat and power heads.
hour	v3/v3.5 surrogate input and TSup controller observation	cyclic sine/cosine	$\sin(2\pi \cdot \text{hour}/24)$; $\cos(2\pi \cdot \text{hour}/24)$	surrogate/rc_node_v35.py envs/tsup_features.py	Avoids discontinuity between hour 23 and hour 0.
day	v3/v3.5 surrogate input and TSup controller observation	cyclic sine/cosine	$\sin(2\pi \cdot \text{day}/365)$; $\cos(2\pi \cdot \text{day}/365)$	surrogate/rc_node_v35.py envs/tsup_features.py	Represents seasonal phase without an artificial year-end jump.

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variable	context	scaling_method	parameters	source_file	justification
a0_raw	direct supply-temperature action	native action domain	$[-1, 1]$ maps to 18..35 C supply temperature	envs/tsup_features.py	Preserves the PPO action coordinate while keeping physical supply-temperature bounds explicit.
a1_raw	direct fan action	native action domain	$[-1, 1]$ maps to 0..1 fan command	envs/tsup_features.py	Keeps fan modulation aligned with the controller action range.
p_total	TSup controller observation, raw mode	affine min-max to $[-1, 1]$	min=0 W; max=5500 W	envs/tsup_features.py	Raw power normalization for ablation baselines.
p_total	canonical TSup controller observation	clipped log transform to $[-1, 1]$	clip at 2500 W; $2 * \log_{10}(p) / \log_{10}(2500) - 1$	envs/tsup_features.py	Compresses high-power tails and improved transfer in the canonical hybrid branch.
delta_t	TSup controller observation, raw mode	linear clipping	clip($\delta_t / 5$ C, -1, 1)	envs/tsup_features.py	Baseline delta-temperature history encoding.
delta_t	MORL 17D TSup-style observation	causal smooth tanh	clip to ± 1.5 C; $\tanh(\delta_t / 1.25)$	envs/tsup_features.py	Reduces action discontinuities from noisy short-step temperature changes.
controller_t_zone	TSup controller observation, raw mode	affine min-max to $[-1, 1]$	min=5 C; max=40 C	envs/tsup_features.py	Wide physical range for controller-side temperature observation.
controller_t_zone	TSup controller observation, comfort-centered ablation	comfort-centered clipping	clip($(T_{\text{zone}} - 22.5 \text{ C}) / 3 \text{ C}$, -1, 1)	envs/tsup_features.py	Ablation tested a comfort-local coordinate, but canonical thermostat hybrid uses raw mode.
power_head_output	v3 surrogate output	Softplus multiplied by P_MAX	P_MAX=5500 W	surrogate/rc_node_v3.py	Guarantees non-negative predicted HVAC power.
C_zon	v3.5 physics backbone	positive reparameterization	$\text{softplus}(\log_{10}(c_{\text{zon}}) - 5) + c_{\text{min}}$ c_min=5e4 J/K	surrogate/rc_node_v3.py	Constrains the identified thermal capacitance to physically meaningful positive values.

Table S8: Training hyperparameters.

Optimiser, learning rate, batch size, regularisation, and stopping criteria.

Table 8: Table S8. Training hyperparameters.

param	value	source_file	justification
active_v3_training_entrypoint	surrogate/train_surrogate_backbone.py	surrogate/train_surrogate_backbone.py	Canonical comfort-oriented direct-TSup v3 backbone training script currently present in active contour.
legacy_train_surrogate_py	not present in active contour	surrogate/train_surrogate.py	The former generic entrypoint is not part of the current active contour; v3 training is reported through train_surrogate_backbone.py.
epochs	500	surrogate/train_surrogate_backbone.py	Default CLI value; upper budget constrained by early stopping.
batch_size	256	surrogate/train_surrogate_backbone.py	Default DataLoader batch size used for stable mini-batch estimates on the BOPTEST corpus.
learning_rate	1e-3	surrogate/train_surrogate_backbone.py	Default AdamW learning rate for the v3 backbone.
hidden_dim	64	surrogate/train_surrogate_backbone.py	Default hidden width used by RCNeuralODEv2 heat and power heads.
optimizer	AdamW	surrogate/train_surrogate_backbone.py	Weight-decayed Adam optimizer stabilizes the black-box residual heads without changing the physical inputs.
weight_decay	1e-4	surrogate/train_surrogate_backbone.py	Explicit optimizer regularization in the canonical v3 training loop.
scheduler	CosineAnnealingLR(T_max=epochs)	surrogate/train_surrogate_backbone.py	Schedule learning-rate decay over the configured epoch budget.
early_stopping	patience=30; improvement_threshold=1e-6	surrogate/train_surrogate_backbone.py	Stops training when the selected validation metric no longer improves.
checkpoint_metric	val_loss default; roll-out_rmse optional	surrogate/train_surrogate_backbone.py	Default keeps one-step supervised fidelity; roll-out_rmse was added for rollout-aware selection when explicitly requested.
gradient_clip_norm	1.0	surrogate/train_surrogate_backbone.py	Enables stable updates from multi-step rollout loss.
multi_horizons	[2, 4]	surrogate/train_surrogate_backbone.py	Default short rollout horizons for one-step-plus-local-rollout training.
loss_lambda_temp	1.0	surrogate/train_surrogate_backbone.py	Phonon temperature fidelity term.
loss_lambda_power	0.1	surrogate/train_surrogate_backbone.py	Phonon power term is normalized by P_MAX and weighted below the temperature term.

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param	value	source_file	justification
loss_lambda_multi	0.5	surrogate/train_surrogate_balances.py	Penalizes local rollout consistency against one-step fit.
loss_lambda_physics	0.05	surrogate/train_surrogate_balances.py	Penalizes implausibly large one-step temperature jumps.

Table S9: Architecture justification.

Layer widths, depths, activations, and ablation evidence.

Table 9: Table S9. Architecture justification.

variant	role	explicit	explicit_block	block	align	ident	24	pol	in	quad	col	erg	phys	peak	whal	gy	trk	typical	temp
v3	control-oriented direct-TSup surrogate	no	no			0.07254147	0.00476512	6329	692393	6822746	827768	2789476	4020137	432362	867				
v35_calibrated_twin	physics-oriented calibrated twin	yes	yes	0.23189610	76249673	44607023	10541160	1929476	6228573	8208569	21493997	3468046	672948	12					
hybrid_l013	13 dynamics regularized by calibrated v3.5	partial	regularizer-only			0.08664903	3674196	362605	727634	1352988	3173604	2710827	661238	98034					

Table S10: Targeted sensitivity analysis.

Per-hyperparameter sweep results, lambda-temp scan included.

Table 10: Table S10. Targeted sensitivity analysis.

sensitivity_axis	tested_values	selection_metric	winner	numerical_reason	positioning
observation_ablation	none,no_power,no_delta_t,no_stat	power	no_delta_t,no_prev_delta_t	Best direct temp history v3.5 diagnostic row was no_delta_t with peak m_s=1.0361 and typical m_s=1.1208.	targeted_sensitivity

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sensitivity_axis	tested_values	selection_metric	winner	numerical_reason	positioning
feature_encoding	raw,causal_smooth,clipped_log,cosine,centered,clipped_log,tEncoding,delta=retargeted	clipped_log,cosine,centered,clipped_log,tEncoding,delta=retargeted	clipped_log,cosine,centered,clipped_log,tEncoding,delta=retargeted	Encoding tests were evaluated through direct v3.5 transfer diagnostics and then reused in the canonical hybrid branch instead of running a global architecture search.	targeted_sensitivity
hybrid_lambda_temp	0.01,0.1,0.15	mean thermo-static m_s across peak/typical	1010	lambda 0.010 achieved the best mean m_s among the tested values (0.0639).	targeted_sensitivity

Table S11: Replicative and predictive validity.

One-step and multi-horizon rollout RMSE across all variants.

Table 11: Table S11. Replicative and predictive validity.

model	horizon	horizon_steps	RMSE_15m	MAE_T	R2_T	RMSE_P	MAE_P	n_windows	power_lambda	lambda_tk
v3	1h	4	1.4709810018239429	0.780517578	1227.559326151873	0.224609125	42848			
				0.9824447164760735						
v3	4h	16	1.5785892009733677	0.7925796509	1218.204588143762	0.2709916091375	169856			
				1.303279318547117						
v3	8h	32	1.5792924165723632	0.687698364	1201.723144857125	0.2481351862	335616			
				1.354566670622206						
v3	24h	96	1.5575184812282702	0.65229415894	1187.580818946825	0.36239976	957696			
				1.4114033281552683						

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model	horizon	horizon_steps	RMSE_L1	MAE_T	R2_T	RMSE_PMAE_P	n_windows	power_lambda	std	std
v3.5_calibrated	4		0.3213366568288526210793292643758128363048009350488128125					42848		
v3.5_calibrated	16		0.575187087052924229436942676767942944147422881787009375					169856		
v3.5_calibrated	32		0.637765884399430826568602615567246369236280231530761819					335616		
v3.5_calibrated	96		0.644130587547873820973442627567520642136271480275978906					957696		
hybrid_l0l1h	4		1.47098100182394298780517578	1227.55932617147302246091275	0.9824447164760735			42848	0.10	5

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model	horizon	horizon_steps	RMSE_15_T	MAE_15_T	R2_15_T	RMSE_PMAE_P	n_windows	power_lambda	power_lambda_tk
hybrid_l010h	16		1.5785892009773677	1.303279318547117	0.925796509	1218.20458814376270991609175	169856	0.10	5
hybrid_l018h	32		1.5792924165773678	1.354566670622206	0.91687698364	1201.72314483712572481354852	335616	0.10	5
hybrid_l0124h	96		1.557518482282706	1.4114033281552683	0.925229415894	1187.5808185946871536239976875	957696	0.10	5

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model	horizon	horizon_steps	RMSE_L5T	MAE_T	R2_T	RMSE_PMAE_P	n_windows	power_lambda	shape	theta
v3_15min_matched	4		0.8745673276257063	0.5561987967			10720			
v3_15min_matched	16		0.8735689528799485	0.579703373948			10624			
v3_15min_matched	32		0.8730632013799723	0.58951713			10496			

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model	horizon	horizon_steps	RMSE_15min	MAE_T	R2_T	RMSE_PMAE_P	n_windows	power_lambda	shape	theta
v3_15min_matched	2h	96	0.8761156751667387	0.6015768552			9984			
raw_v35	1h	4	1.4908158751731406	0.623571486			10720			
raw_v35	4h	16	1.4912848911222379	0.980305441			10624			

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model	horizon	horizon_steps	RMSE_15_T	MAE_15_T	R2_15_T	RMSE_PMAE_P	n_windows	power_lambda	stepda_tk
raw_v35	8h	32	1.4907779021275325	1.235325	0.7005360918		10496		
raw_v35	24h	96	1.4664688641205789	1.205789	0.4030921147		9984		

Compliance summary

Of the seventeen Hou-and-Evins Reporting-Level-3 items, fifteen are covered with quantitative evidence by tables S1–S11. The two uncovered items are (i) physical co-simulation between two independent Modelica engines (only one engine is used here through the BOPTEST FMU) and (ii) grid-tied multi-agent coordination (out of scope for the single-zone testbed used in the main study).

Audit-anchor cross-references

- MORL canonical selection pre-registration: commit 93df9b3.
- MORL post- $N=5$ canonical falsification: commit 62dc859.
- Block 3 transferability pre-registration: manifest field `audit.pre_registration_commit_sha` in `configs/block3_testcase_manifest.yaml`.
- Block 3 closure: manifest field `audit.block3_close_commit_sha`.